Math 54 — Week 9

Inner product, length and orthogonality; orthogonal sets; orthogonal projections (Lay 6.1–6.3)

§1 Facts used from Weeks 1–8

(W1) **Matrix–vector products.** $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$, the combination of the columns of A with weights the entries of $\mathbf{x}$ (Lay 1.4). Entry i of $A\mathbf{x}$ is row i of A times $\mathbf{x}$, and entry (i,j) of $A^\mathsf{T}B$ is column i of A times column j of B. Also $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$.

(W2) **The subspaces of a matrix.** For an $m \times n$ matrix A: $\operatorname{Nul} A$ is the set of solutions of $A\mathbf{x} = \mathbf{0}$, $\operatorname{Col} A$ is the span of the columns, and $\operatorname{Row} A$ is the span of the rows. $\dim \operatorname{Row} A = \dim \operatorname{Col} A = \operatorname{rank} A$, and $\operatorname{rank} A + \dim \operatorname{Nul} A = n$ (rank theorem, Lay 4.5–4.6).

(W3) **Bases.** A set of p linearly independent vectors in a p-dimensional subspace is a basis. Coordinates relative to a basis are unique.

(W4) **Invertible matrix theorem.** If A is square and $BA = I$ for some B, then A is invertible and $A^{-1} = B$.

(W5) **Determinants.** $\det(AB) = \det A \det B$ and $\det A^\mathsf{T} = \det A$ (Lay 3.2).

§2 This week

§2.1 Inner product, length, distance (Lay 6.1)

The question. In the plane, the arrow $(3,4)$ has length 5 by the Pythagorean theorem, and $(1,2)$ is perpendicular to $(-2,1)$. What should the length of $(1,-2,2,0) \in \mathbb{R}^4$ be, and when should two vectors of $\mathbb{R}^4$ count as perpendicular?

Definition. The **inner product** (dot product) of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^\mathsf{T}\mathbf{v} = u_1v_1 + \cdots + u_nv_n.$$

The **length** (norm) is $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$, and the **distance** between $\mathbf{u}$ and $\mathbf{v}$ is $\|\mathbf{u} - \mathbf{v}\|$. A **unit vector** has length 1.

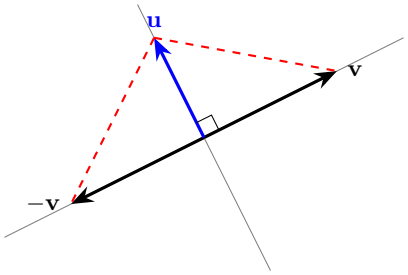
- **Rules.** Each follows in one line from the formula: $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$; $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$; $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v})$; $\mathbf{u} \cdot \mathbf{u} = u_1^2 + \cdots + u_n^2 \geq 0$, with equality only for $\mathbf{u} = \mathbf{0}$. From the third rule, $\|c\mathbf{v}\| = \sqrt{c^2\,\mathbf{v} \cdot \mathbf{v}} = |c|\,\|\mathbf{v}\|$.
- **Why this is length.** In $\mathbb{R}^2$, $\|(a,b)\| = \sqrt{a^2+b^2}$ is the length of the arrow by the Pythagorean theorem. In $\mathbb{R}^3$, the arrow (a,b,c) is the hypotenuse of a right triangle whose horizontal leg is $(a,b,0)$, of length $\sqrt{a^2+b^2}$, and whose vertical leg has length $|c|$; so its length is $\sqrt{a^2+b^2+c^2}$. In $\mathbb{R}^n$ we take the same formula as the definition.
- **Normalizing.** For $\mathbf{v} \neq \mathbf{0}$, the vector $\mathbf{v}/\|\mathbf{v}\|$ has length $\frac{1}{\|\mathbf{v}\|}\|\mathbf{v}\| = 1$; it is the unit vector in the direction of $\mathbf{v}$. Example: $\|(1,-2,2,0)\| = \sqrt{1+4+4} = 3$, so $\frac{1}{3}(1,-2,2,0)$ is a unit vector.

- **Examples.** $(2,-5,-1)\cdot(3,2,-3) = 6-10+3 = -1$. The distance between $(7,1)$ and $(3,2)$ is $\|(4,-1)\| = \sqrt{17}$.

Perpendicular lines. In the plane, take $\mathbf{v} \neq \mathbf{0}$. The segment from $-\mathbf{v}$ to $\mathbf{v}$ has midpoint $\mathbf{0}$, so the line through $\mathbf{0}$ perpendicular to $\mathbf{v}$ is the perpendicular bisector of that segment. A point lies on the perpendicular bisector of a segment exactly when it is equally far from the two ends. So the line through $\mathbf{0}$ and $\mathbf{u}$ is perpendicular to $\mathbf{v}$ exactly when $\|\mathbf{u}-\mathbf{v}\| = \|\mathbf{u}-(-\mathbf{v})\|$. Expand both squared distances with the rules:

$$\|\mathbf{u}-\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2\mathbf{u}\cdot\mathbf{v} + \|\mathbf{v}\|^2, \qquad \|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\mathbf{u}\cdot\mathbf{v} + \|\mathbf{v}\|^2.$$

The two are equal exactly when $\mathbf{u}\cdot\mathbf{v} = 0$, and that condition makes sense in every $\mathbb{R}^n$.



$\mathbf{u}$ lies on the perpendicular bisector of the segment from $-\mathbf{v}$ to $\mathbf{v}$ exactly when the two dashed distances are equal. Expanding the two squared distances shows that this happens exactly when $\mathbf{u}\cdot\mathbf{v} = 0$.

Definition. $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if $\mathbf{u}\cdot\mathbf{v} = 0$. ($\mathbf{0}$ is orthogonal to every vector.)

So in the question, $(1,2)\cdot(-2,1) = 0$, and the length of $(1,-2,2,0)$ is 3.

Theorem (Pythagorean theorem in $\mathbb{R}^n$)
$\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Proof. Expand with the rules:

$$\|\mathbf{u}+\mathbf{v}\|^2 = (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v}) = \|\mathbf{u}\|^2 + 2\,\mathbf{u}\cdot\mathbf{v} + \|\mathbf{v}\|^2.$$

This equals $\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ exactly when $\mathbf{u}\cdot\mathbf{v} = 0$. □

Angles in $\mathbb{R}^2$ and $\mathbb{R}^3$. The law of cosines for the triangle with sides $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u}-\mathbf{v}$ says $\|\mathbf{u}-\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta$, where θ is the angle between the arrows. Comparing with the expansion above gives

$$\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta.$$

Shortcuts

- **Unit vector**: divide by the length. Reason: $\|c\mathbf{v}\| = |c|\,\|\mathbf{v}\|$ with $c = 1/\|\mathbf{v}\|$. Condition: $\mathbf{v} \neq \mathbf{0}$. Instance: $(2,0,-1)$ has length $\sqrt{5}$, so $\frac{1}{\sqrt{5}}(2,0,-1)$ is a unit vector.
- **Compare lengths by their squares**: $\|\mathbf{u}\| \le \|\mathbf{v}\|$ exactly when $\|\mathbf{u}\|^2 \le \|\mathbf{v}\|^2$, because both lengths are nonnegative. Instance: $(1,1,1)$ is shorter than $(2,0,0)$ because $3 < 4$; no square roots are needed.
- **The sign of $\mathbf{u}\cdot\mathbf{v}$ gives the type of angle** in $\mathbb{R}^2$ and $\mathbb{R}^3$: positive means acute, 0 right, negative obtuse, because $\|\mathbf{u}\|\,\|\mathbf{v}\| > 0$ for nonzero vectors. Instance: $(2,-5,-1)\cdot(3,2,-3) = -1$, so the angle is obtuse.

Traps

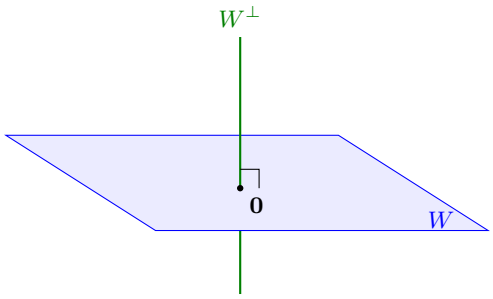
- $\|\mathbf{u}+\mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$ is false in general: for $\mathbf{u} = (1,0)$, $\mathbf{v} = (0,1)$ the left side is $\sqrt{2}$ and the right side is 2. The Pythagorean theorem is about squares.
- $\|c\mathbf{v}\| = c\|\mathbf{v}\|$ is false for $c < 0$: $\|-2(1,0)\| = 2$, not -2. The correct rule has $|c|$.

§2.2 Orthogonal complements (Lay 6.1)

The question. Which vectors $\mathbf{x} \in \mathbb{R}^3$ are orthogonal to both $(1,1,1)$ and $(1,-1,0)$, and therefore to every vector of the plane they span? What kind of set do they form, and how do we compute it?

Definition. If W is a subspace of $\mathbb{R}^n$, its **orthogonal complement** $W^\perp$ is the set of vectors orthogonal to every vector of W.

- **Testing membership.** If $W = \operatorname{Span}\{\mathbf{w}_1, \dots, \mathbf{w}_p\}$, then $\mathbf{x} \in W^\perp$ if and only if $\mathbf{x}$ is orthogonal to each $\mathbf{w}_j$. Reason: if $\mathbf{x} \cdot \mathbf{w}_j = 0$ for all j, then by the rules
$$\mathbf{x} \cdot (c_1\mathbf{w}_1 + \cdots + c_p\mathbf{w}_p) = c_1(\mathbf{x} \cdot \mathbf{w}_1) + \cdots + c_p(\mathbf{x} \cdot \mathbf{w}_p) = 0.$$
- **$W^\perp$ is a subspace.** It contains $\mathbf{0}$, and if $\mathbf{x}, \mathbf{y} \in W^\perp$, then for every $\mathbf{w} \in W$, $(\mathbf{x} + c\mathbf{y}) \cdot \mathbf{w} = 0 + c \cdot 0 = 0$.
- **$W \cap W^\perp = \{\mathbf{0}\}$.** A vector $\mathbf{v}$ in both is orthogonal to itself, so $\mathbf{v} \cdot \mathbf{v} = 0$ and $\mathbf{v} = \mathbf{0}$.



In $\mathbb{R}^3$, the complement of a plane W through $\mathbf{0}$ is the line through $\mathbf{0}$ perpendicular to it, and the complement of that line is the plane.

Theorem (Row, null and column spaces (Lay Thm. 3))
For an $m \times n$ matrix A: $(\operatorname{Row} A)^\perp = \operatorname{Nul} A$ and $(\operatorname{Col} A)^\perp = \operatorname{Nul} A^\mathsf{T}$.

The idea

$A\mathbf{x} = \mathbf{0}$ says, entry by entry, that each row of A dotted with $\mathbf{x}$ is 0 (W1). That is exactly the statement that $\mathbf{x}$ is orthogonal to every row, hence to the span of the rows. The second statement is the first applied to A^T, whose rows are the columns of A.

Proof. 1. By (W1), entry i of $A\mathbf{x}$ is (row i of A)$\cdot\mathbf{x}$. So $A\mathbf{x} = \mathbf{0}$ if and only if $\mathbf{x}$ is orthogonal to every row of A.

2. By the membership test, that holds if and only if $\mathbf{x} \in (\operatorname{Row} A)^\perp$. So $\operatorname{Nul} A = (\operatorname{Row} A)^\perp$.

3. The rows of A^T are the columns of A, so $\operatorname{Row} A^\mathsf{T} = \operatorname{Col} A$. Applying step 2 to A^T gives $\operatorname{Nul} A^\mathsf{T} = (\operatorname{Col} A)^\perp$.

□

Computing $W^\perp$. Put spanning vectors of W as the *rows* of a matrix A; then $W = \operatorname{Row} A$ and $W^\perp = \operatorname{Nul} A$. For the question,
$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \end{bmatrix},$$
so $x_1 = -\frac{1}{2}x_3$ and $x_2 = -\frac{1}{2}x_3$ with x_3 free. Taking $x_3 = -2$, $W^\perp$ is the line through $(1,1,-2)$. Check: $(1,1,1) \cdot (1,1,-2) = 0$ and $(1,-1,0) \cdot (1,1,-2) = 0$.

Dimensions. With $W = \operatorname{Row} A$ as above, $\dim W = \operatorname{rank} A$ and $\dim W^\perp = \dim \operatorname{Nul} A$, so the rank theorem (W2) gives
$$\dim W + \dim W^\perp = n.$$
In the example, $2 + 1 = 3$. Every vector of W is orthogonal to every vector of $W^\perp$, so $W \subseteq (W^\perp)^\perp$. Applying the dimension formula to $W^\perp$ gives $\dim(W^\perp)^\perp = n - \dim W^\perp = \dim W$, so $(W^\perp)^\perp = W$ (W3).

Shortcuts

- $W^\perp$ **from a spanning set**: spanning vectors as rows, then the null space. Check $\dim W + \dim W^\perp = n$ and one dot product per pair. Instance: the computation above.
- W **given by one equation**: if $W = \{\mathbf{x} : \mathbf{a} \cdot \mathbf{x} = 0\}$ with $\mathbf{a} \neq \mathbf{0}$, then $W = (\mathrm{Span}\{\mathbf{a}\})^\perp$, so $W^\perp = \mathrm{Span}\{\mathbf{a}\}$ because $(V^\perp)^\perp = V$. Instance: the plane $x_1 + x_2 - x_3 = 0$ in $\mathbb{R}^3$ has complement the line through $(1, 1, -1)$.

Traps

- The complement of $\mathrm{Col}\, A$ is $\mathrm{Nul}\, A^\mathsf{T}$, not $\mathrm{Nul}\, A$. For
$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix},$$
$\mathrm{Col}\, A = \mathrm{Span}\{(1, 0)\}$ and $\mathrm{Nul}\, A = \mathrm{Span}\{(1, -1)\}$, but $(1, -1) \cdot (1, 0) = 1 \neq 0$. The complement is $\mathrm{Nul}\, A^\mathsf{T} = \mathrm{Span}\{(0, 1)\}$.
- Being orthogonal to one vector of W does not put $\mathbf{x}$ in $W^\perp$; test every vector of a spanning set. For W the x_1x_2-plane in $\mathbb{R}^3$, $\mathbf{x} = (0, 1, 0)$ is orthogonal to $(1, 0, 0) \in W$ but lies in W.

§2.3 Orthogonal sets (Lay 6.2)

The question. Let $\mathbf{u}_1 = (3, 1, 1)$, $\mathbf{u}_2 = (-1, 2, 1)$, $\mathbf{u}_3 = (-\frac{1}{2}, -2, \frac{7}{2})$ and $\mathbf{y} = (6, 1, -8)$. Find the weights c_1, c_2, c_3 with $\mathbf{y} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3$ without row reducing.

Definition. $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an **orthogonal set** if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ whenever $i \neq j$. An **orthogonal basis** of W is a basis that is an orthogonal set.

Theorem (Orthogonal sets of nonzero vectors are independent (Lay Thm. 4))
If $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors, it is linearly independent, and so it is a basis of the subspace it spans.

The idea

To show a weight c_j in a relation $c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p = \mathbf{0}$ is 0, take the dot product of the relation with $\mathbf{u}_j$. Orthogonality makes every term except the j-th equal to 0.

Proof. 1. Suppose $c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p = \mathbf{0}$. Take the dot product of both sides with $\mathbf{u}_j$:
$$c_1(\mathbf{u}_1 \cdot \mathbf{u}_j) + \cdots + c_p(\mathbf{u}_p \cdot \mathbf{u}_j) = \mathbf{0} \cdot \mathbf{u}_j = 0.$$
2. Every term with $i \neq j$ is 0, so $c_j(\mathbf{u}_j \cdot \mathbf{u}_j) = 0$. Since $\mathbf{u}_j \neq \mathbf{0}$, $\mathbf{u}_j \cdot \mathbf{u}_j > 0$, so $c_j = 0$.
3. This holds for each j, so the set is independent, and by definition it is a basis of its span.

□

Theorem (Weights in an orthogonal basis (Lay Thm. 5))
If $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an orthogonal basis of W, then each $\mathbf{y} \in W$ is
$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p, \qquad c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}.$$

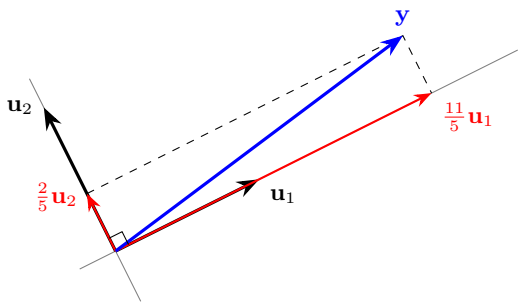
Proof. 1. Since the $\mathbf{u}_j$ form a basis of W, there are weights with $\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p$.

2. Take the dot product with $\mathbf{u}_j$. Because $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ for $i \neq j$,

$$\mathbf{y} \cdot \mathbf{u}_j = c_1(\mathbf{u}_1 \cdot \mathbf{u}_j) + \cdots + c_p(\mathbf{u}_p \cdot \mathbf{u}_j) = c_j(\mathbf{u}_j \cdot \mathbf{u}_j).$$

3. Divide by $\mathbf{u}_j \cdot \mathbf{u}_j$, which is not 0 because $\mathbf{u}_j \neq \mathbf{0}$.

□



$\mathbf{y} = (4, 3)$ in the orthogonal basis $\mathbf{u}_1 = (2, 1)$, $\mathbf{u}_2 = (-1, 2)$: the weights are $c_1 = \frac{8+3}{5} = \frac{11}{5}$ and $c_2 = \frac{-4+6}{5} = \frac{2}{5}$. Each term $c_j\mathbf{u}_j$ is the foot of the perpendicular from $\mathbf{y}$ to the line through $\mathbf{u}_j$ (dashed).

The question answered. The three dot products are

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = -3 + 2 + 1 = 0, \qquad \mathbf{u}_1 \cdot \mathbf{u}_3 = -\frac{3}{2} - 2 + \frac{7}{2} = 0, \qquad \mathbf{u}_2 \cdot \mathbf{u}_3 = \frac{1}{2} - 4 + \frac{7}{2} = 0.$$

So the $\mathbf{u}_j$ are nonzero and orthogonal, hence independent (Thm. 4); three independent vectors in $\mathbb{R}^3$ are a basis (W3). By Thm. 5,

$$c_1 = \frac{18 + 1 - 8}{11} = 1, \qquad c_2 = \frac{-6 + 2 - 8}{6} = -2, \qquad c_3 = \frac{-3 - 2 - 28}{33/2} = -2,$$

so $\mathbf{y} = \mathbf{u}_1 - 2\mathbf{u}_2 - 2\mathbf{u}_3$.

Shortcuts

- **Weights in an orthogonal basis**: one quotient of dot products each, $c_j = (\mathbf{y} \cdot \mathbf{u}_j)/(\mathbf{u}_j \cdot \mathbf{u}_j)$. Condition: the basis is orthogonal. Instance: the question above.
- **Orthogonal basis of $\mathbb{R}^n$**: n nonzero vectors that are pairwise orthogonal form a basis, with no row reduction; check the $n(n-1)/2$ dot products. Reason: Thm. 4 and (W3). Instance: the three dot products above.

Traps

- An orthogonal set may contain $\mathbf{0}$, and then it is dependent: $\{\mathbf{0}, (1, 0)\}$ is orthogonal. Thm. 4 needs nonzero vectors.
- The weight formula needs an orthogonal basis. With the basis $(1, 0)$, $(1, 1)$ of $\mathbb{R}^2$ and $\mathbf{y} = (0, 1)$, the formula gives $c_1 = 0$ and $c_2 = \frac{1}{2}$, but $0(1, 0) + \frac{1}{2}(1, 1) = (\frac{1}{2}, \frac{1}{2}) \neq \mathbf{y}$.

§2.4 Projection onto a line (Lay 6.2)

The question. Write $\mathbf{y} = (7, 6)$ as a multiple of $\mathbf{u} = (4, 2)$ plus a vector perpendicular to $\mathbf{u}$. Which point of the line through $\mathbf{u}$ is closest to $\mathbf{y}$, and how far away is it?

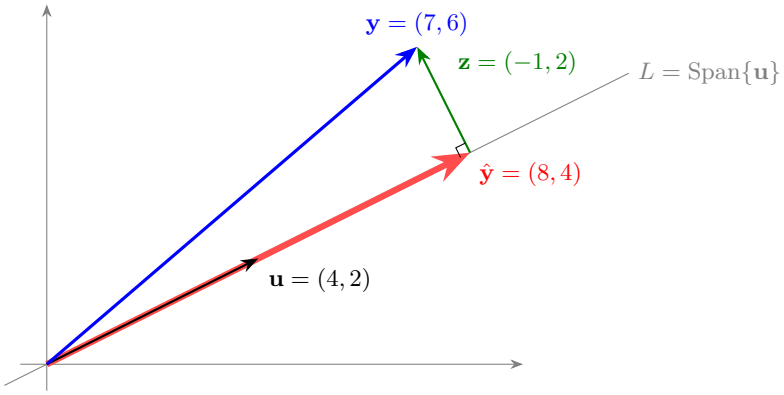
We want $\mathbf{y} = \alpha\mathbf{u} + \mathbf{z}$ with $\mathbf{z} \cdot \mathbf{u} = 0$. Substituting $\mathbf{z} = \mathbf{y} - \alpha\mathbf{u}$, the condition is $\mathbf{y} \cdot \mathbf{u} - \alpha\, \mathbf{u} \cdot \mathbf{u} = 0$, so $\alpha = (\mathbf{y} \cdot \mathbf{u})/(\mathbf{u} \cdot \mathbf{u})$. With $L = \text{Span}\{\mathbf{u}\}$:

$$\hat{\mathbf{y}} = \text{proj}_L\, \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}\, \mathbf{u}, \qquad \mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}.$$

For any other point $\beta\mathbf{u}$ of L, the vectors $\mathbf{z}$ and $\hat{\mathbf{y}} - \beta\mathbf{u}$ are orthogonal ($\mathbf{z}$ is orthogonal to every multiple of $\mathbf{u}$), so by the Pythagorean theorem

$$\|\mathbf{y} - \beta\mathbf{u}\|^2 = \|\mathbf{z}\|^2 + \|\hat{\mathbf{y}} - \beta\mathbf{u}\|^2 \geq \|\mathbf{z}\|^2.$$

So $\hat{\mathbf{y}}$ is the closest point of L, and the distance from $\mathbf{y}$ to L is $\|\mathbf{z}\|$. In Thm. 5, each term $c_j\mathbf{u}_j$ is the projection of $\mathbf{y}$ onto $\text{Span}\{\mathbf{u}_j\}$: an orthogonal basis splits $\mathbf{y}$ into its projections onto the lines through the basis vectors.



The question answered: $\mathbf{y} \cdot \mathbf{u} = 40$ and $\mathbf{u} \cdot \mathbf{u} = 20$, so $\hat{\mathbf{y}} = 2\mathbf{u} = (8, 4)$ (red, under $\mathbf{u}$), and $\mathbf{z} = (-1, 2)$ is orthogonal to $\mathbf{u}$. The distance from $\mathbf{y}$ to L is $\|\mathbf{z}\| = \sqrt{5}$.

Shortcuts

- **No need to normalize $\mathbf{u}$**: replacing $\mathbf{u}$ by $c\mathbf{u}$ ($c \neq 0$) multiplies the numerator and the denominator of α by c and c^2, and the product $\alpha\mathbf{u}$ is unchanged. Instance: $\mathbf{u} = (2, 1)$ gives $\frac{20}{5}(2, 1) = (8, 4)$, the same $\hat{\mathbf{y}}$.
- **Check**: $\mathbf{z} \cdot \mathbf{u}$ must be 0. Instance: $(-1, 2) \cdot (4, 2) = 0$.

Traps

- $\text{proj}_L\, \mathbf{y}$ is a vector; the number $(\mathbf{y} \cdot \mathbf{u})/(\mathbf{u} \cdot \mathbf{u})$ is only its coefficient. In the example the coefficient is 2 and the projection is $(8, 4)$.
- The distance to L is $\|\mathbf{y} - \hat{\mathbf{y}}\|$, not $\|\hat{\mathbf{y}}\|$: here $\sqrt{5}$, not $\sqrt{80}$.

§2.5 Orthonormal sets and orthogonal matrices (Lay 6.2)

The question. The columns $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ and $\mathbf{u}_2 = \frac{1}{3}(2, -2, 1)$ of a 3×2 matrix U are orthogonal unit vectors. For $\mathbf{x} = (\sqrt{2}, 3)$, compare $\|U\mathbf{x}\|$ with $\|\mathbf{x}\|$.

Definition. An **orthonormal set** is an orthogonal set of unit vectors. An **orthonormal basis** of W is a basis that is an orthonormal set.

- **Normalizing an orthogonal set.** If $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are nonzero and orthogonal, then the vectors $\mathbf{v}_j/\|\mathbf{v}_j\|$ are unit vectors, and for $i \neq j$ their dot product is $(\mathbf{v}_i \cdot \mathbf{v}_j)/(\|\mathbf{v}_i\|\, \|\mathbf{v}_j\|) = 0$. So they form an orthonormal set.
- **Weights in an orthonormal basis.** The denominators in Thm. 5 are $\mathbf{u}_j \cdot \mathbf{u}_j = 1$, so $c_j = \mathbf{y} \cdot \mathbf{u}_j$.

Theorem (Orthonormal columns (Lay Thms. 6 and 7))
An $m \times n$ matrix U has orthonormal columns if and only if $U^\mathsf{T}U = I$. In that case, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}, \qquad \|U\mathbf{x}\| = \|\mathbf{x}\|,$$

and $U\mathbf{x}$, $U\mathbf{y}$ are orthogonal exactly when $\mathbf{x}$, $\mathbf{y}$ are.

Proof. 1. By (W1), entry (i, j) of $U^\mathsf{T}U$ is $\mathbf{u}_i \cdot \mathbf{u}_j$. The columns are orthonormal exactly when this is 1 for $i = j$ and 0 for $i \neq j$, that is, when $U^\mathsf{T}U = I$.

2. If $U^\mathsf{T}U = I$, then
$$(U\mathbf{x}) \cdot (U\mathbf{y}) = (U\mathbf{x})^\mathsf{T}(U\mathbf{y}) = \mathbf{x}^\mathsf{T}U^\mathsf{T}U\mathbf{y} = \mathbf{x}^\mathsf{T}\mathbf{y} = \mathbf{x} \cdot \mathbf{y}.$$

3. Taking $\mathbf{y} = \mathbf{x}$ gives $\|U\mathbf{x}\|^2 = \|\mathbf{x}\|^2$. The statement about orthogonality is the case $\mathbf{x} \cdot \mathbf{y} = 0$.

□

The question answered. By (W1),

$$U\mathbf{x} = \sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, 1, 0) + 3 \cdot \frac{1}{3}(2, -2, 1) = (3, -1, 1),$$

so $\|U\mathbf{x}\| = \sqrt{11}$, and $\|\mathbf{x}\| = \sqrt{2 + 9} = \sqrt{11}$, as the theorem says.

Orthogonal matrices. A *square* matrix with orthonormal columns is an **orthogonal matrix**. Then $U^\mathsf{T}U = I$, so $U^{-1} = U^\mathsf{T}$ by (W4). Example: the rotation matrix

$$U = \begin{bmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{bmatrix}$$

has columns of length $\sqrt{\cos^2 t + \sin^2 t} = 1$ whose dot product is $-\cos t \sin t + \sin t \cos t = 0$, so it is orthogonal, and its inverse U^T is the rotation by $-t$.

Shortcuts

- **Test for orthonormal columns**: compute $U^\mathsf{T}U$; all the dot products of columns appear at once. Instance: for the U of the question, $\mathbf{u}_1 \cdot \mathbf{u}_2 = \frac{1}{3\sqrt{2}}(2 - 2 + 0) = 0$ and both columns have length 1, so $U^\mathsf{T}U = I_2$.
- **Inverse of an orthogonal matrix**: $U^{-1} = U^\mathsf{T}$, no row reduction. Reason: (W4). Instance: the rotation above.
- **Lengths after multiplying by** U: $\|U\mathbf{x}\| = \|\mathbf{x}\|$, so compute the length of $\mathbf{x}$ instead. Instance: the question above.

Traps

- An orthogonal matrix has orthonormal columns, not just orthogonal ones. The matrix
$$U = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$
has orthogonal columns, but $U^\mathsf{T}U = 2I$, so it is not an orthogonal matrix, and its inverse is $\frac{1}{2}U^\mathsf{T}$, not U^T.

§2.6 Orthogonal projections (Lay 6.3)

The question. $\mathbf{u}_1 = (2, 5, -1)$ and $\mathbf{u}_2 = (-2, 1, 1)$ span a plane W in $\mathbb{R}^3$, and $\mathbf{y} = (1, 2, 3)$. Which point of W is closest to $\mathbf{y}$, and how far is $\mathbf{y}$ from W?

For a line, we subtracted the projection onto the line. For a plane with an orthogonal basis, try subtracting the projections onto both basis lines.

Theorem (Orthogonal decomposition theorem (Lay Thm. 8))

Let W be a subspace of $\mathbb{R}^n$ with an orthogonal basis $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$. Each $\mathbf{y} \in \mathbb{R}^n$ can be written uniquely as $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$ with $\hat{\mathbf{y}} \in W$ and $\mathbf{z} \in W^\perp$, and

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 + \cdots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p}\mathbf{u}_p.$$

$\hat{\mathbf{y}}$ is the **orthogonal projection of $\mathbf{y}$ onto W**, written $\text{proj}_W \mathbf{y}$.

The idea

Project $\mathbf{y}$ onto each line $\text{Span}\{\mathbf{u}_j\}$ separately and add. Because the $\mathbf{u}_j$ are mutually orthogonal, subtracting all the projections leaves a vector orthogonal to each $\mathbf{u}_j$, hence to all of W. Uniqueness: two such decompositions would give a vector lying in both W and $W^\perp$, which is $\mathbf{0}$.

Proof. 1. *Existence.* Define $\hat{\mathbf{y}}$ by the formula; it is a combination of the $\mathbf{u}_j$, so $\hat{\mathbf{y}} \in W$. Let $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

2. For each j, take the dot product of $\mathbf{z}$ with $\mathbf{u}_j$. In $\hat{\mathbf{y}} \cdot \mathbf{u}_j$ every term except the j-th is 0, so

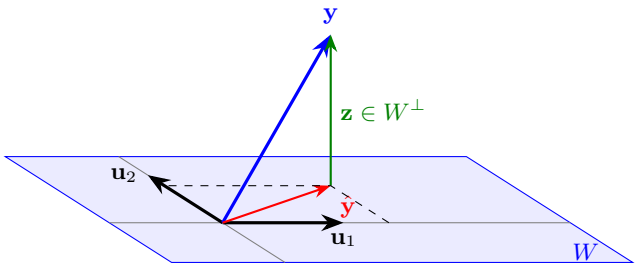
$$\mathbf{z} \cdot \mathbf{u}_j = \mathbf{y} \cdot \mathbf{u}_j - \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}(\mathbf{u}_j \cdot \mathbf{u}_j) = 0.$$

   So $\mathbf{z}$ is orthogonal to every $\mathbf{u}_j$, hence to every vector of W (membership test): $\mathbf{z} \in W^\perp$.

3. *Uniqueness.* If also $\mathbf{y} = \hat{\mathbf{y}}_1 + \mathbf{z}_1$ with $\hat{\mathbf{y}}_1 \in W$ and $\mathbf{z}_1 \in W^\perp$, then $\hat{\mathbf{y}} - \hat{\mathbf{y}}_1 = \mathbf{z}_1 - \mathbf{z}$. The left side is in W and the right side is in $W^\perp$, so this vector is in $W \cap W^\perp = \{\mathbf{0}\}$. Hence $\hat{\mathbf{y}} = \hat{\mathbf{y}}_1$ and $\mathbf{z} = \mathbf{z}_1$.

4. In particular $\hat{\mathbf{y}}$ does not depend on which orthogonal basis of W is used: two bases would give two decompositions $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$ of this kind, and step 3 says there is only one.

□



$\hat{\mathbf{y}} = \text{proj}_W \mathbf{y}$ is the sum of the projections of $\mathbf{y}$ onto the lines through $\mathbf{u}_1$ and $\mathbf{u}_2$ (gray); the dashed segments join those two projections to $\hat{\mathbf{y}}$. The vector $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$ is perpendicular to the plane.

The question, first part. $\mathbf{u}_1 \cdot \mathbf{u}_2 = -4 + 5 - 1 = 0$, so the basis is orthogonal. With $\mathbf{y} \cdot \mathbf{u}_1 = 9$, $\mathbf{u}_1 \cdot \mathbf{u}_1 = 30$, $\mathbf{y} \cdot \mathbf{u}_2 = 3$ and $\mathbf{u}_2 \cdot \mathbf{u}_2 = 6$:

$$\hat{\mathbf{y}} = \frac{9}{30}\mathbf{u}_1 + \frac{3}{6}\mathbf{u}_2 = \left(-\frac{2}{5},\ 2,\ \frac{1}{5}\right), \qquad \mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \left(\frac{7}{5},\ 0,\ \frac{14}{5}\right).$$

Check that $\mathbf{z}$ is orthogonal to the basis:

$$\mathbf{z} \cdot \mathbf{u}_1 = \frac{14}{5} - \frac{14}{5} = 0, \qquad \mathbf{z} \cdot \mathbf{u}_2 = -\frac{14}{5} + \frac{14}{5} = 0.$$

If $\mathbf{y}$ is already in W, then $\mathbf{y} = \mathbf{y}+\mathbf{0}$ is a decomposition of the required kind, so by uniqueness $\text{proj}_W \mathbf{y} = \mathbf{y}$.

Theorem (Best approximation theorem (Lay Thm. 9))

Let W be a subspace of $\mathbb{R}^n$ with an orthogonal basis, and let $\hat{\mathbf{y}} = \operatorname{proj}_W \mathbf{y}$. Then $\hat{\mathbf{y}}$ is the point of W closest to $\mathbf{y}$: $\|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\|$ for every $\mathbf{v} \in W$ with $\mathbf{v} \neq \hat{\mathbf{y}}$.

The idea

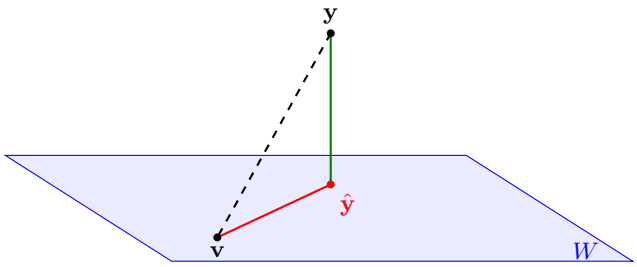
Go from $\mathbf{y}$ to any $\mathbf{v} \in W$ in two perpendicular legs: first from $\mathbf{y}$ to $\hat{\mathbf{y}}$ (a vector in $W^\perp$), then within W from $\hat{\mathbf{y}}$ to $\mathbf{v}$. By the Pythagorean theorem the direct path is longer than the first leg unless the second leg is zero.

Proof. 1. Write $\mathbf{y} - \mathbf{v} = (\mathbf{y} - \hat{\mathbf{y}}) + (\hat{\mathbf{y}} - \mathbf{v})$. The first vector is $\mathbf{z} \in W^\perp$; the second is in W (a difference of vectors in W). So they are orthogonal.

2. By the Pythagorean theorem,
$$\|\mathbf{y} - \mathbf{v}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \mathbf{v}\|^2.$$

3. If $\mathbf{v} \neq \hat{\mathbf{y}}$, the last term is positive, so $\|\mathbf{y} - \mathbf{v}\| > \|\mathbf{y} - \hat{\mathbf{y}}\|$.

□



Best approximation: $\mathbf{y} - \mathbf{v}$ (dashed) is the hypotenuse of a right triangle with legs $\mathbf{y} - \hat{\mathbf{y}}$ (green, perpendicular to W) and $\hat{\mathbf{y}} - \mathbf{v}$ (red, in W), so it is longer than $\mathbf{y} - \hat{\mathbf{y}}$.

The question, second part. The closest point of W to $\mathbf{y}$ is $\hat{\mathbf{y}} = (-\frac{2}{5}, 2, \frac{1}{5})$, and the distance from $\mathbf{y}$ to W is

$$\|\mathbf{y} - \hat{\mathbf{y}}\| = \left\|\left(\frac{7}{5}, 0, \frac{14}{5}\right)\right\| = \frac{7}{5}\sqrt{1+4} = \frac{7\sqrt{5}}{5}.$$

Theorem (The projection with an orthonormal basis (Lay Thm. 10))

If $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an orthonormal basis of W and $U = [\,\mathbf{u}_1 \cdots \mathbf{u}_p\,]$, then

$$\operatorname{proj}_W \mathbf{y} = (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p = UU^{\mathsf{T}}\mathbf{y}.$$

Proof. 1. The denominators $\mathbf{u}_j \cdot \mathbf{u}_j$ in Thm. 8 are 1, which gives the first formula.

2. By (W1), $U^{\mathsf{T}}\mathbf{y}$ is the vector of dot products $(\mathbf{u}_1 \cdot \mathbf{y}, \ldots, \mathbf{u}_p \cdot \mathbf{y})$, and U times a vector is the combination of the columns of U with the entries as weights. So $UU^{\mathsf{T}}\mathbf{y}$ is the first formula.

□

Example. Take $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,0,1)$, $\mathbf{u}_2 = (0,1,0)$ and $\mathbf{y} = (1,2,3)$. Then

$$U = \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & 1 \\ 1/\sqrt{2} & 0 \end{bmatrix}, \qquad U^{\mathsf{T}}\mathbf{y} = \begin{bmatrix} 4/\sqrt{2} \\ 2 \end{bmatrix} = \begin{bmatrix} 2\sqrt{2} \\ 2 \end{bmatrix}, \qquad UU^{\mathsf{T}}\mathbf{y} = 2\sqrt{2}\,\mathbf{u}_1 + 2\mathbf{u}_2 = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}.$$

The matrix of the projection is

$$UU^{\mathsf{T}} = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{bmatrix}.$$

Shortcuts

- **Check a projection**: $\mathbf{y} - \hat{\mathbf{y}}$ must be orthogonal to each basis vector of W. Instance: the checks in the first part of the question.
- **Project onto the complement when it is a line**: if $W^\perp = \mathrm{Span}\{\mathbf{n}\}$, then
$$\mathrm{proj}_W \mathbf{y} = \mathbf{y} - \frac{\mathbf{y}\cdot\mathbf{n}}{\mathbf{n}\cdot\mathbf{n}}\mathbf{n}.$$
Reason: $\mathbf{y} = \mathbf{z} + \hat{\mathbf{y}}$ with $\mathbf{z} \in W^\perp$ and $\hat{\mathbf{y}} \in W = (W^\perp)^\perp$, so by uniqueness for the line $W^\perp$, $\mathbf{z}$ is the projection of $\mathbf{y}$ onto that line. Instance: for the plane $x_1 + x_2 + x_3 = 0$ and $\mathbf{y} = (1,2,3)$, $\mathbf{n} = (1,1,1)$ and $\mathrm{proj}_W \mathbf{y} = (1,2,3) - 2(1,1,1) = (-1,0,1)$.
- **Part of an orthogonal basis of $\mathbb{R}^n$**: if $\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_n\mathbf{u}_n$ in an orthogonal basis and $W = \mathrm{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$, then $\mathrm{proj}_W \mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k$. Reason: the remaining terms are in $W^\perp$, and the decomposition is unique. Instance: in the orthogonal-sets question, $\mathbf{y} = \mathbf{u}_1 - 2\mathbf{u}_2 - 2\mathbf{u}_3$, so the projection onto $\mathrm{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$ is $\mathbf{u}_1 - 2\mathbf{u}_2 = (5,-3,-1)$.
- **Distance to W**: $\|\mathbf{y} - \mathrm{proj}_W \mathbf{y}\|$.

Traps

- The projection formula needs an orthogonal basis. With the basis $(1,0)$, $(1,1)$ of $W = \mathbb{R}^2$ and $\mathbf{y} = (0,1)$, the sum of the two line projections is $(0,0) + (\frac{1}{2}, \frac{1}{2}) \neq \mathbf{y}$, although $\mathrm{proj}_W \mathbf{y} = \mathbf{y}$.
- The closest point is $\hat{\mathbf{y}}$; the distance is $\|\mathbf{y} - \hat{\mathbf{y}}\|$. In the question, $\|\hat{\mathbf{y}}\| = \frac{\sqrt{105}}{5}$ is not the distance $\frac{7\sqrt{5}}{5}$.

§3 Homework this week

HW8, Section 6.1: 1, 7, 9, 13, 17, 22, 23, 27, 30, 34. Section 6.2: 5, 7, 9, 15, 17, 23, 24, 26, 27, 29, 32. Section 6.3: 3, 7, 11, 15, 17, 22, 28, 30. (Section 6.4, also on the course calendar this week, is in the Week 10 handout.)

§4 Problem set

P1. For $\mathbf{u} = (1,2,-1,3)$ and $\mathbf{v} = (2,-1,3,1)$, compute $\mathbf{u}\cdot\mathbf{v}$, $\|\mathbf{u}\|$, $\|\mathbf{v}\|$ and $\|\mathbf{u}+\mathbf{v}\|$. Are $\mathbf{u}$ and $\mathbf{v}$ orthogonal? Check the Pythagorean theorem for them.

P2. Find a basis of $W^\perp$ for $W = \mathrm{Span}\{(1,0,1,1),(0,1,-1,2)\} \subseteq \mathbb{R}^4$.

P3. Show that $\mathbf{b}_1 = (1,1,0)$, $\mathbf{b}_2 = (1,-1,1)$, $\mathbf{b}_3 = (-1,1,2)$ form an orthogonal basis of $\mathbb{R}^3$ and write $\mathbf{x} = (3,1,4)$ in this basis without row reduction.

P4. Write $\mathbf{y} = (3,1)$ as the sum of a vector on the line through $\mathbf{u} = (1,2)$ and a vector orthogonal to $\mathbf{u}$. What is the distance from $\mathbf{y}$ to the line?

P5. Let $W = \mathrm{Span}\{(1,1,0),(1,-1,1)\}$ and $\mathbf{y} = (2,3,4)$. Find $\mathrm{proj}_W \mathbf{y}$, the distance from $\mathbf{y}$ to W, and the closest point of W to $\mathbf{y}$.

P6. Let A be $m \times n$. Show that if $A^\mathsf{T}A\mathbf{x} = \mathbf{0}$ then $A\mathbf{x} = \mathbf{0}$. (Compute $\|A\mathbf{x}\|^2$.) Conclude that $\mathrm{Nul}\, A^\mathsf{T}A = \mathrm{Nul}\, A$.

P7. Let U and V be $n \times n$ orthogonal matrices. Show that $\det U = \pm 1$ and that UV is an orthogonal matrix.

P8. Let $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,0,1)$, $\mathbf{u}_2 = (0,1,0)$, $U = [\,\mathbf{u}_1\ \mathbf{u}_2\,]$, $W = \mathrm{Col}\, U$ and $P = UU^\mathsf{T}$. Show that $P^2 = P$ and $P^\mathsf{T} = P$, and find $\mathrm{Nul}\, P$.